Register Number: 20101200 7

Name of the Candidate: Balaji .G

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B.E. DEGREE EXAMINATION, 2026

(COMMON to CS, IT and AI & ML)

(FOURTH SEMESTER)

22CSBS - 401/ ITBS-401/AIBS-401: DISCRETE MATHEMATICS

May /June] [Time : 3 Hours

Maximum : 75 Marks

PART- A (10 x 1 = 10 Marks)
Answer ALL Questions

		Mar	CO	PO	BTL
1.	Define Dual of a logical statement	1	CO1	1,2	K1
2.	Show that $\neg P \Rightarrow (P \rightarrow Q)$	1	CO1	2,3	K2
3.	State De Morgan's laws of set theory	1	CO2	1,2	K1
4.	Define composition of relations	1	CO2	1,2	K1
5.	Define a distributive Lattice	1	CO3	2,3	K1
6.	State idempotent laws of Boolean algebra	1	CO3	1,2	K1
7.	Define a Monoid with an example	1	CO4	1,2	K1
8.	What is the identity element of the algebraic system $\{S,*\}$ Where S is the set of integers and * is defined by $a * b = a + b + 2$ for all $a, b \in S$.	1	CO4	3,4	K2
9.	Define regular graph with an example	1	CO5	1,2	K1
10.	Define graph isomorphism	1	CO5	1,2	K1

PART- B (5 x 3 = 15 Marks)
Answer either (a) or (b) in each Question

			Mar	CO	PO	BTL
11.	(a)	Prove that $(p \rightarrow q) \wedge (r \rightarrow q) \equiv (p \vee r) \rightarrow q$	3	CO1	1,2	K2
		(OR)				
	(b)	Prove that $p \rightarrow (q \rightarrow p) \equiv \neg p \rightarrow (p \rightarrow q)$	3	CO1	2,3	K2
12.	(a)	If A,B and C are any three sets, Prove that $(A - B) - C = (A - C) - (B - C)$	3	CO2	2,3	K2
		(OR)				
	(b)	If $R = \{(1,1),(1,2),(2,1),(2,2),(3,3),(4,4),(4,5),(5,4),(5,5),(6,6)\}$ is the equivalence relation on the set $A = \{1,2,3,4,5,6\}$ Find the partition of A induced by R.	3	CO2	2,3	K2
13.	(a)	Draw the Hasse diagram for the poset $\{S_{30}, D\}$. Hence prove that it is a lattice	3	CO3	1,2	K2
		(OR)				
	(b)	Simplify the Boolean expression $a'.b'.c + a.b'.c + a'.b'.c'$	3	CO3	2,3	K2
14.	(a)	Show that in a group G, $(a * b)^{-1} = b^{-1} * a^{-1}$, for all $a, b \in G$ (OR)	3	CO4	1,2	K2
	(b)	Show that $\{(1,2,3,4,5,6), \times_7\}$ is a cyclic group.	3	CO4	3,4	K2
15.	(a)	Explain Konigsberg bridge problem. Represent the problem by means of graph. Does the problem have a solution?	3	CO5	1,2	K2
		(OR)				

(b) Define Bipartite graph. Draw $K_{2,3}$ and $K_{3,3}$ graphs. | 3 | CO5 | 2,3 | K2 |

PART- C (4 x10 = 40 Marks)
Answer either (a) or (b) in each Question

16. (a) Obtain the PDNF of $(R \vee \neg P) \rightarrow \neg(Q \rightarrow P)$ | 10 | CO1 | 2,3 | K3 |

(OR)

(b) Show that $\neg p$ is logically follows from the premises $(p \rightarrow q) \wedge (r \rightarrow s)$, $(q \rightarrow t) \wedge (s \rightarrow u)$, $\neg(t \wedge u)$ and $p \rightarrow r$ | 10 | CO1 | 3,4 | K3 |

17. (a) Show that the relation R " congruent modulo 3" is an equivalence relation on the set of integers | 10 | CO2 | 2,3 | K3 |

(OR)

(b) Show that for any three sets $A \times (B \cap C) = (A \times B) \cap (A \times C)$ | 10 | CO2 | 1,3 | K3 |

18. (a) Show that in a Boolean algebra the complement of every element is unique | 10 | CO3 | 1,2 | K3 |

(OR)

(b) Simplify the Boolean function $f(a, b, c, d) = \Sigma(0,1,2,3,4,6,7,8,9,11,15)$ | 10 | CO3 | 2,3 | K3 |

19. (a) Show that $\{S_3, *\}$, where $S = \{1,2,3\}$ is a group under the operation of right composition of permutation | 10 | CO4 | 2,3 | K3 |

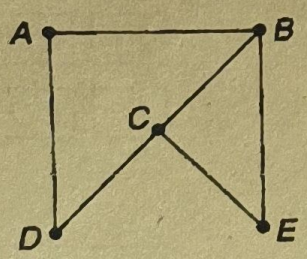
(OR)

(b) List the code words generated by the encoding function | 10 | CO4 | 2,3 | K3 |

$e: B^2 \rightarrow B^6$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

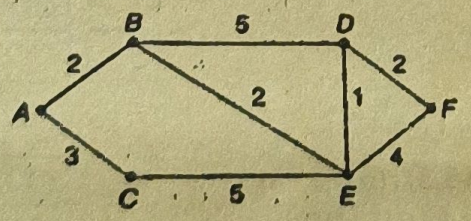
PART- D (1 x 10 = 10 Marks)
Answer either (a) or (b)

20. (a) Summarize all the paths of length 4 between D and E for the following graph using adjacency matrix. | 10 | CO5 | 2,3 | K4 |



(OR)

(b) Find the shortest path from the vertex A to the vertex F | 10 | CO5 | 2,3 | K4 |



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